

EDUGENIE — DATA FLOW DIAGRAM

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LEVEL 0 (Context Diagram)

[Student] → (Question / Topic / Text) → [EduGenie System]

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[Student] ← (Answer / Explanation / Quiz / Learning Path)

LEVEL 1 (Main Components)

Student (browser at <http://127.0.0.1:8000>)

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index.html form

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[Five parallel routes]

└— GET /qa?question=... → Answer

└— POST /explain {topic} → Explanation

└— POST /summarize {text} → Summary

└— POST /quiz {text} → JSON MCQs

└— GET /learn/recommendations?topic=... → Learning Path

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FastAPI backend (app/main.py)

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Google Gemini API (gemini-3.8-flash)

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JSON response back to browser



User sees result in output box

DATA STORES

- .env file → stores **GOOGLE_API_KEY**
- No database needed (stateless design)

KEY FUNCTION CALLS

- **_gen(prompt)** → universal Gemini caller
- **answer_question(q)**
- **explain_topic(topic)**
- **summarize_text(text)**
- **generate_quiz(text)** → parses JSON from Gemini
- **learning_path(topic)**